

Quantification of Depletion-Induced Adhesion of Red Blood Cells

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Red blood cells (RBCs) are known to form aggregates in the form of rouleaux due to the presence of plasma proteins under physiological conditions. The formation of rouleaux can also be induced *in vitro* by the addition of macromolecules to the RBC suspension. Current data on the adhesion strength between red blood cells in their natural discocyte shapes mostly originate from indirect measurements such as flow chamber experiments, but data is lacking at the single cell level. Here, we present measurements on the dextran-induced aggregation of red blood cells using atomic force microscopy-based single cell force spectroscopy. The effects of dextran concentration and molecular weight on the interaction energy of adhering RBCs were determined. The results on adhesion energy are in excellent agreement with a model based on the depletion effect and previous experimental studies. Furthermore, our method allowed to determine the adhesion force, a quantity that is needed in theoretical investigations on blood flow.

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Nonpathological aggregation of red blood cells (RBCs) or “rouleaux formation” [Fig. 1(a)] *in vivo* is frequently observed and caused by the fibrinogen in plasma [1]. These aggregates are reversibly formed and can be dispersed by moderate shear rates. Thus, the shear thinning viscosity of blood is determined to a large extent by the formation and breaking of these aggregates. Irreversible red blood cell aggregation could be a microcirculatory risk factor and indicative of disease because irreversible aggregates can be observed in diseases such as malaria, multiple myeloma, inflammation [2], or pathological thrombus formation [3]. Reversible rouleaux formation can also be induced by resuspending the RBCs in physiological solutions containing neutral macromolecules such as dextran (DEX) [4]. The fibrinogen-mediated aggregation of RBCs increases consistently with increasing fibrinogen concentration [5], whereas the dextran-mediated aggregation of RBCs seems to reach a maximum at a certain dextran concentration. The strength of the aggregation depends not only on the dextran concentration but also on the molecular weight of the dextran (i.e., the radius of gyration of the dextran) [6,7]. Previously, two different models were developed to explain the aggregation of RBCs in polymer solutions: the cross-bridging model and the depletion model. The bridging model has been favored for a long time and has been proposed for plasma protein and neutral dextran macromolecule-induced RBC aggregation [8]. In this model, it is assumed that fibrinogen or dextran molecules nonspecifically adsorb onto the cell membrane and form a “bridge” to the adjacent cell. However, over the most recent decade, more and more evidence has been observed in favor of the depletion model [9–13]. A first explanation of depletion forces was given by Asakura and Oosawa [14], who discovered that the presence of small spheres (i.e., macromolecules) can induce effective forces between two

larger particles if the distance between them is small enough. The origin of these forces is purely entropic. If the distance between the two large particles decreases to less than the size of the surrounding macromolecules, these macromolecules are expelled from the region between the particles. Consistently, the concentration of macromolecules becomes depleted in this region compared to that of the bulk, and an effective osmotic pressure causing an attraction between the large particles occurs. Neu and Meiselman [9] applied this concept of depletion-induced adhesion to red blood cells and developed a theoretical description of the interaction energy between two red blood cells in terms of the molecular weight and concentration of the dextran used. These results were compared with former measurements reported by Buxbaum *et al.*

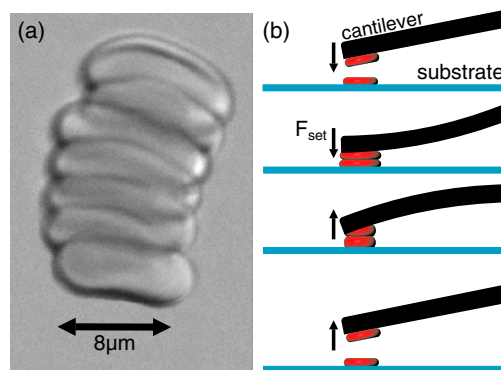


FIG. 1 (color online). (a) Snapshot of a rouleau of seven RBCs in a dextran solution. (b) A sketch of the working principle of SCFS. A single cell is attached to the prefunctionalized cantilever and is lowered onto another cell, which is fixed at the bottom. Both cells are pushed together with an appropriate set point force F_{set} . The adhesion force and adhesion energy are measured while withdrawing the cells.

[15] based on the micropipette aspiration technique. Neu *et al.* were able to adapt their model parameters consistently to the experimental data of Buxbaum *et al.*, but the number of data points remained limited.

In the present study, we used the technique of single cell force spectroscopy (SCFS) [Fig. 1(b)] to measure the interaction energies between human red blood cells as functions of the molecular weight and concentration of dextran at the single cellular level and compared them to the predicted values of Neu and Meiselman [9]. The dextrans used were dextran70 (DEX70 with a molecular weight of 70 kDa) and dextran150 (DEX150 with a molecular weight of 150 kDa) from Sigma-Aldrich. In SCFS, a single RBC is attached to a cantilever, while another cell is attached to the surface of the Petri dish. Both cells are now brought into close contact, and during the withdrawing of the cantilever the adhesion force between both cells is measured via the deflection of the cantilever [see Fig. 1(b)]. An atomic force microscope (AFM) (Nanowizard 2, equipped with the CellHesion module with an increased pulling range of 100 μm , JPK Instruments, Germany) was used to conduct single cell force spectroscopy measurements [16]. The spring constant of the cantilever was determined using the common thermal noise method (the cantilevers used were MLCT-O cantilevers with spring constants on the order of 0.01 N/m, Bruker). Fresh blood from healthy donors was taken using a finger prick (three different donors). The blood was obtained within 1 day of the experiment and was stored at room temperature during and between measurements. The total time between finger prick and the end of the experiments did not exceed 5 h (average time 2 h). The cells were washed three times by centrifugation (800 g, 3 min) in a phosphate-buffered solution of physiological ionic strength. In the course of the experiment, a single RBC was attached to an AFM cantilever by appropriate functionalization. Cell-Tak (BD Science) was used to bind a cell to the cantilever. Cell-Tak is a strong tissue adhesive consisting of marine mussel proteins. The binding is of nonspecific nature and is assumed not to activate any cells. A protocol was used, in which the cantilever was incubated in a Cell-Tak drop. After 2 min, the Cell Tak solution was carefully removed; this was followed by a 3 min waiting period to allow the acetic acid from the Cell-Tak to evaporate from the cantilever. Rinsing the cantilever with ethanol and phosphate-buffered saline (PBS, 137 mM/1NaCl, 2.7mM/1KCl, 10 mM/1Na₂HPO₄·2H₂O, 2 mM/1KH₂PO₄, pH = 7.4) completed the functionalization protocol. The attachment procedure took place in a PBS solution with a certain amount of dextran present. To attach a RBC to the cantilever, the latter was lowered manually until a preset cantilever deflection (i.e., the force set point F_{set}) was reached, indicating contact between the cantilever and the cell. Afterwards, the cantilever was withdrawn continuously at low speed until the cell was no longer in contact with the surface. At this point, it is essential that the adhesion of the cell to the Cell-Tak is greater than the adhesion of the cell to

the Petri dish surface. The adhesion of the top cell to the cantilever is stronger than the adhesion of a cell to the surface. However, the adhesion among two cells is smaller than both the adhesion of the top cell to the cantilever and the bottom cell to the surface. After the cell was attached to the cantilever, the cell was placed above another cell that laid on the bottom of the Petri dish. Accordingly, the cantilever (with attached cell) was lowered until the two cells came into close contact. Functionalization of the plastic Petri dish (Polystyrol), to immobilize the bottom cell, was not necessary because RBCs adhered to the surface without any further treatment. While withdrawing (or retracting) the cantilever, the adhesion force and adhesion energy were measured. The retraction curve is typically characterized by the maximum force required to separate the cells from each other, and adhesion energy is calculated by computing the area under the retraction curve of the force distance curve. The interaction energy densities of two RBCs are calculated by dividing the measured adhesion energies by the contact areas of the adhering cells using a value of 50.24 μm^2 derived from the maximum radius of RBCs. The present study is exclusively concerned with adhesion that is caused by the presence of dextran molecules in the solution. Hence, any further source of adhesion [e.g., adhesion of the lower RBC to the Cell-Tak-coated surface of the cantilever, see Fig. 2(a)] had to be excluded. For larger and stiffer cells, compared to RBCs, such undesired adhesion events are rarely observed. For RBCs that have a height of just 2 μm , this is an experimental difficulty; even with optimum (concentric) alignment, such binding to the surface was often observed in our first experiments. An example of those undesired adhesion events is shown in Fig. 2(a). The measured adhesion forces were much higher than any reasonable estimate for dextran-induced adhesion. To overcome this problem, 0.1 g/dl bovine serum albumin (BSA) was added to the solution after attaching the RBC to the cantilever by pipetting several drops of BSA to the solution. The concentrations of dextran and BSA were carefully calculated to adjust the correct concentrations of dextran. The radius of gyration of BSA is only 3 nm [17], and the concentration is fairly small so that any additional depletion interactions due to the presence of BSA can be neglected. The effects of BSA treatment on RBCs (e.g., on cell geometry or mechanical properties) have been studied intensively [18–20], and any effect of the BSA treatment on the measured interaction energy can be excluded because the investigated adhesion is supposed to be purely physical (i.e., we assume that no adhesion proteins are involved there that could possibly be blocked by the incorporation of BSA into the RBC membrane). In agreement with the literature, we found that in most of the cases, both the cells on the Petri dish as well as on the cantilever remained in their physiological discocyte shape. Echinocytic cells have not been measured. The only purpose of the BSA is to passivate the surfaces of the cantilever and the Petri dish. Thus, only the RBC surfaces contribute to the measured adhesion force arising from the

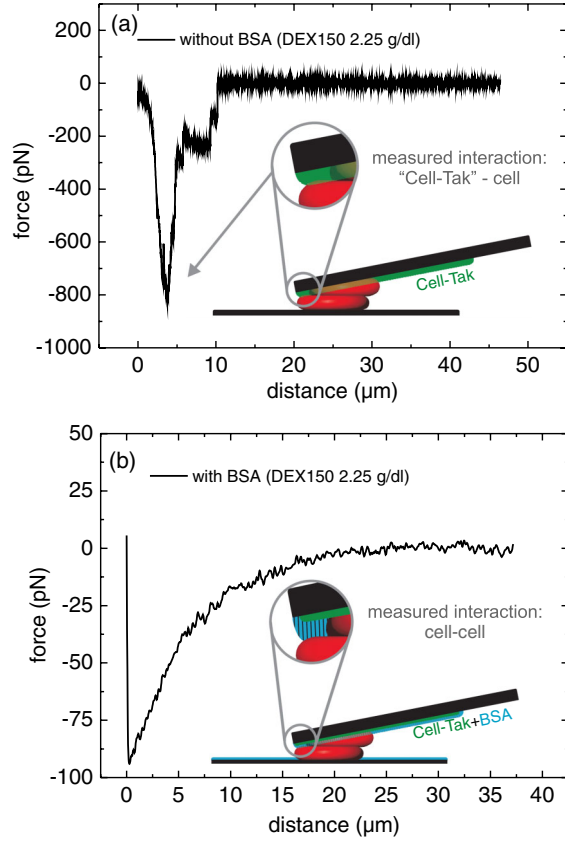


FIG. 2 (color online). [(a) and (b)] Effects of BSA treatment (see the text for details). Without BSA treatment, undesired adhesion events occur whose origin is not the investigated depletion effect; e.g., the cells do not hit concentrically, and the lower cell touches the Cell-Tak (i.e., a stronger adhesion force is measured because of the strong adhesiveness of the Cell-Tak). With BSA treatment, the Cell-Tak is completely passivated and the influence of those undesired adhesion events is minimized, as the changes in shape and magnitude of the measured force curve document.

depletion effect. Figure 2(b) shows an example force curve after BSA treatment of the cantilever and the Petri dish. The shape and magnitude of the force distance curve change significantly.

In the next step, the parameter set point force F_{set} , cantilever velocity v , and contact time τ_{set} of the cantilever were adjusted. Figure 3(a) shows the interaction energy as a function of F_{set} for DEX70, DEX150, and a measurement without dextran (control). The influence of the choice of F_{set} on the measured interaction energy is negligible and was set to $F_{\text{set}} = 300$ pN for the remaining measurements. A small interaction energy can be measured in the control measurements (without dextran). This small force possibly arises as a side effect of the squeezing of the cells during the approach. Due to this squeezing, there can be interactions events between the two glycocalyxes. The absolute value, however, is small compared to those derived in the measurements with dextran. We will show below that we

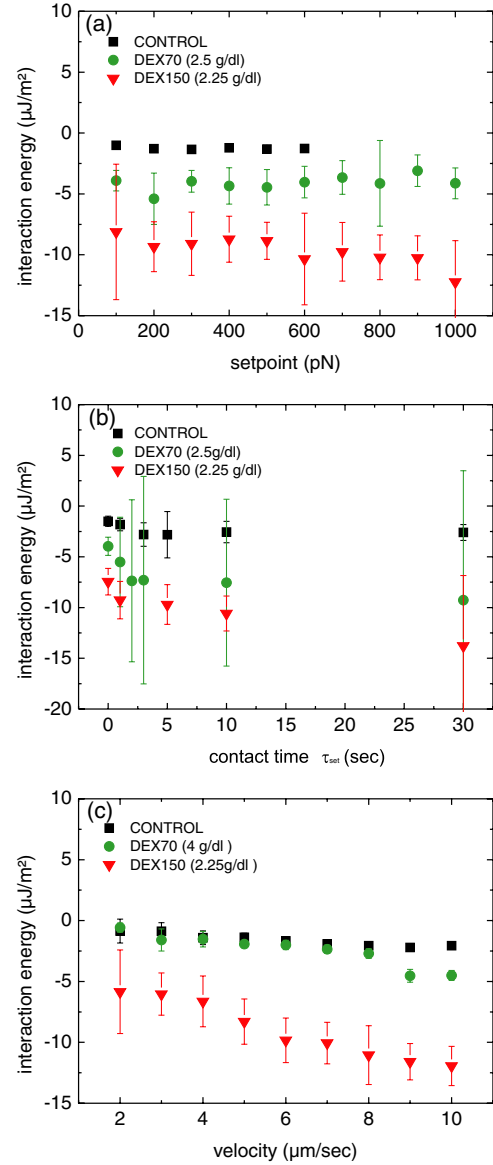


FIG. 3 (color online). Role of the parameters. (a) Dependence of the measured adhesion energy on the chosen force set point F_{set} . In all measurements, no significant dependence on F_{set} was observed. (b) Dependence of the measured adhesion energy on the chosen contact time τ_{set} of both cells. Increasing contact time leads to an increase in interaction energy and error bars (SD). (c) Dependence of the measured adhesion energy on the chosen withdrawal velocity v of the cantilever. At higher velocities in the DEX150 measurements, a dependence on the cantilever velocity was observed, but for moderate velocities this dependence was still less than the error measurement.

can describe our data well with a depletion model, but it is also known that macromolecular “bridging” between the RBCs can occur when the cells are in contact for a longer time [21]. Bronkhorst *et al.* [21] discovered that the time constant for those possible cross bridges is on the order of seconds. We varied the contact time between first contact of the two cells (when F_{set} was reached) and start of the

retraction of the cantilever from $\tau_{\text{set}} = 0$ s to $\tau_{\text{set}} = 30$ s [Fig. 3(b)]. Large contact times lead to slightly increased adhesion energies and increased error bars. Both could be indications for bridging events. Therefore, we minimized the contact time by setting τ_{set} in the control software to 0 s. Of course the contact time during which the cells touch before the cantilever retracts will be slightly larger since it takes approximately 0.2 s after first contact to reach the preset force set point F_{set} . Furthermore, depending on the cantilever velocity, the RBCs will still be in contact for some time while the cantilever retracts. Due to their flexibility, the RBCs can stay in contact over distances of several μm [Fig. 2(b)]. The ability to deform quite extensively has already been seen in previous studies [22,23]. The total deformation of the cell depends on the interaction strength (i.e., the concentration of dextran used). According to the amount of deformation, the contact time of the cells over the whole measurement τ_{total} varied between less than 1 and 4 s. Hence, to minimize the actual contact time (to exclude bridging effects), the cantilever velocity had to be sufficiently high. This ensures that the measured interaction energies are purely depletion-induced. Figure 3(c) shows the dependence of the cantilever velocity v on the measured interaction energies. In the control measurements, no influence of the velocity could be seen. On the contrary, a dependence of the velocity could be observed in the dextran measurements. We do not have a conclusive explanation for this dependence, but we assume that higher interaction energies lead to larger viscoelastic effects while deforming the RBCs. Due to the higher velocity, the RBCs are deformed to a greater extent, and this might lead to higher apparent interaction energies. Additionally, at the given size and concentration of dextran70 and dextran150, the buffer has a large increase in viscosity. Combined with the size of the cantilever, this provides a simple explanation for the observed velocity dependence. For dextran70, the effect is smaller since the viscosity depends on the molecular weight and concentration. Up to a certain velocity, the effect can be neglected; e.g., for dextran70 this effect begins at velocities only higher than $9 \mu\text{m/s}$. For larger values of v , in the dextran150 measurements, this effect can be significant; for moderate velocities, this effect is still less than the error measurement. On the other side, the v should not be too small to exclude any bridging effects. Therefore, in all measurements, the cantilever velocity was chosen as $v = 5 \mu\text{m/s}$.

Figure 4 shows the dependence of the adhesion force and the interaction energy on the dextran concentration. Each data point represents an average of approximately 100 force curves for different cells of different donors. Between a single concentration, 5–10 cells were tested. The cell at the cantilever was changed for each new concentration. The measured interaction energies are in excellent agreement with the predicted interaction energies

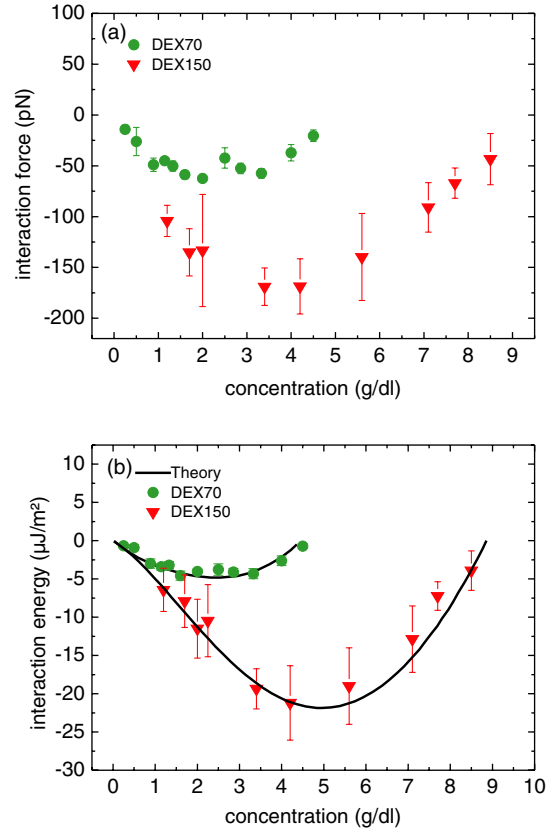


FIG. 4 (color online). (a) Measured adhesion forces of the dextrans in terms of concentrations. The maximum interaction strengths were observed at 2 g/dl (DEX70) and 4 g/dl (DEX150). (b) The dependence of the interaction energy of two RBCs on the concentrations of two dextran types. The solid line represents the curve calculated by Neu and Meiselman [9]. The errors are given in SD. The error bars in the DEX70 measurements are smaller than the symbol size.

given by Neu and Meiselman [9], who used an analytical approach to calculate the depletion interaction energy E_D between two RBCs [9],

$$E_D = -2\pi(\Delta - d/2 + \delta - p), \quad (1)$$

where π is the osmotic pressure, Δ is thickness of the depletion layer, d is the separation distance between adjacent surfaces, δ is the glycocalyx thickness, and p is the depth of polymer penetration into the glycocalyx. Their model combines electrostatic repulsion due to RBC surface charge and osmotic attractive forces due to polymer depletion near the RBC surface. The theory considers the soft surfaces of RBCs and the subsequent penetration depth p of polymers into the surface. This penetration depth p depends on the polymer type, concentration, and molecular size and is expected to be greater for small molecules and to increase with increasing polymer concentration due to increasing osmotic pressure π . With increasing osmotic pressure, the penetration of macromolecules into the soft RBCs deepens, impeding the depletion of macromolecules

between both cells and hence reducing the interaction energy. Above a threshold concentration this effect becomes dominant, and this decreases the interaction energy, even though the concentration of macromolecules is increasing further. Taking the depletion effect and the soft surfaces of the RBCs into account, a bell-shaped dependence of the interaction energy on the dextran concentration was calculated [the solid line in Fig. 4(b)]. Our approach also allows the simultaneous determination of adhesion energies and adhesion forces. The adhesion force according to the dextran concentration is shown in Fig. 4(a). This force is in the range of 14–23 pN for dextran 70 and 43–169 pN for dextran 150. In the literature, no experimental data is available on the depletion-induced adhesion forces between single RBCs.

In conclusion, we have presented single-cell force spectroscopy measurements on dextran-induced red blood cell aggregation. The presence of dextran mimics the plasma molecules that lead to the formation of rouleaux under physiological conditions. Our findings of the interaction energy are in excellent agreement with a previous study [9], and they can be described by a model based on the depletion effect. Furthermore, we present the first direct experimental quantification of dextran-induced depletion forces between RBCs in their natural discocyte shape. For contact times longer than a few seconds, we find a slight tendency towards stronger adhesion energies. We can not conclusively decide whether this is due to bridging of the macromolecules between two cells or due to some other effect. One would need a side view [24] of the adhesion areas while performing the adhesion tests to analyze this effect in greater detail. Additionally, the investigation of the occurring adhesion using a more physiological depletion agent such as fibrinogen has to be done in order to check for any bridging effects.

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